

Rare Earth Or Rare Ingenuity? India Remains Between The Two

With China firmly controlling rare earth exports, India confronts a critical moment in its technological trajectory. The challenge is no longer limited to mineral access; it lies in innovation itself. The key test is whether Indian startups, scientists, and policymakers can reduce reliance on rare earths through magnet-free motor designs and advanced materials development.



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Following the successful closure of the Electronics Component Manufacturing Scheme application window on September 30, 2025, Union Minister for Electronics and IT Ashwini Vaishnaw stated that India's electronics manufacturing sector faces no bottlenecks in the domestic supply of rare earth minerals. It was quite a claim amid the buzz, given that rare earth vulnerabilities have dominated industry discussions throughout the year.

Over the last four decades, China has built near-total dominance in rare earth processing and magnet manufacturing through abundant resources, state-backed investment, and vertical integration. From early mining in the 1980s to full control of refining and magnet production by the 2000s, China's coordinated policies, low costs, and industrial clustering created global dependency. Export curbs in 2010 triggered geopolitical alarm, while recent export licensing has revived concerns about the supply chain. Today, China controls an estimated 70-95% of global rare earth processing, prompting diversification drives in the US, EU, Japan, Australia, and India.

Back in April 2025, when China added a new chapter to its tightening of rare earth exports, India's electronics, electric vehicle (EV), and defence sectors felt the pinch, exposing vulnerabilities in critical supply chains.

India's rare earth roadmap: Between sovereignty and supply chains

India had recognised the issue much earlier, elevating rare earths from a materials concern to a national security and technology priority. As Principal Scientific Adviser to the government of India, Ajay Kumar Sood emphasised, the national goal is "to make the country self-sufficient or *Aatmanirbhar* in this critical area."

Sood explained that while rare earth elements "are not actually rare," the challenge lies in their "extraction, which is highly technology driven." To address this, India Rare Earths Limited (IREL), operating under the Department of Atomic Energy, has taken the lead in extraction and processing. IREL "extracts rare earth oxides from monazite sands, a process that has been ongoing for many years."